Part 3: Gradient Descent Manual Calculation

$y = mx + b$ ⇒ (linear model)

$$\left.\begin{aligned} m_0 &= -1 \\ b_0 &= 1 \end{aligned}\right\} \text{Initial Value}$$

Learning rate $\alpha = 0.1$

$(1,3), (3,6)$ → Data point

And number of points

$n = 2$

<u>Cost Function (MSE)</u>

$$J(m,b) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

Prediction ⇒ $\hat{y} = mx + b$

Gradients

$$\frac{\partial J}{\partial m} = \frac{2}{n}\sum(\hat{y} - y)x$$

$$\frac{\partial J}{\partial b} = \frac{2}{n}\sum(\hat{y}-y)$$

When come to update rule

$$m = m - \alpha \frac{\partial J}{\partial m}$$

$$b = b - \alpha \frac{\partial J}{\partial b}$$

Starting Iteration 1

$m_0 = -1, \; b_0 = 1$

$$\left.\begin{aligned} \hat{y}_1 &= (-1)(1)+1 = 0 \\ \hat{y}_2 &= (-1)(3)+1 = -2 \end{aligned}\right\} \text{Prediction}$$

$$\left.\begin{aligned} e_1 &= 0-3 = -3 \\ e_2 &= -2-6 = -8 \end{aligned}\right\} \text{Errors}$$

So Since $\frac{2}{n} = \frac{2}{2} = 1$

Gradient for m $\rightarrow \frac{\partial J}{\partial m} = 1\left[(e_1)(x_1)+(e_2)(x_2)\right]$

$$\frac{\partial J}{\partial m} = (-3)(1)+(-8)(3)$$

$$= -3-24$$

$$= -27$$

Gradient for b $\Rightarrow \frac{\partial J}{\partial m} = -1(e_1 + e_2)$

$$\frac{\partial J}{\partial b} = -3 + (-8)$$
$$= -11$$

Then the update

$$m_1 = -1 - 0.1(-27)$$
$$= -1 + 2.7 = 1.7$$

$$b_1 = 1 - 0.1(-11)$$
$$= 1 + 1.1$$
$$= 2.1$$

Result from Iteration 1

$m_1 = 1.7$, $b_1 = 2.1$

CamScanner

Iteration 2

$m_1 = 1.7$, $b_1 = 2.1$

$$\left.\begin{aligned} \hat{y}_1 &= 1.7(1) + 2.1 \\ &= 3.8 \end{aligned}\right\} \text{Prediction}$$

$$\left.\begin{aligned} \hat{y}_2 &= 1.7(3) + (2.1) \\ &= 5.1 + 2.1 \\ &= 7.2 \end{aligned}\right\} \text{Prediction}$$

$$\left.\begin{aligned} e_1 &= 3.8 - 3 \\ &= 0.8 \end{aligned}\right\} \text{error}$$

$$\left.\begin{aligned} e_2 &= 7.2 - 6 \\ &\cancel{= 1.6} = 1.2 \end{aligned}\right\} \text{error}$$

CamScanner

Date / /

Gradient for m

$$\frac{\partial J}{\partial m} = (0.8)(1) + (1.2)(3)$$

$$= 0.8 + 3.6$$

$$= 4.4$$

Gradient for b

$$\frac{\partial J}{\partial b} = 0.8 + 1.2$$

$$= 2.0$$

The update

$$m_2 = 1.7 - 0.1(4.4)$$

$$= 1.7 - 0.44$$

$$= 1.26$$

$$b_2 = 2.1 - 0.1(2.0)$$

$$= 2.1 - 0.2$$

$$= 1.9$$

Date / /

Per the calculations on Iteration 2

$m_2 = 1.26$, $b_2 = 1.9$

CamScanner

Iteration 3

$m_2 = 1.26, \; b_2 = 1.9$

$$\left.\begin{aligned} \hat{y}_1 &= 1.26(1) + 1.9 = 3.16 \\ \hat{y}_2 &= 1.26(3) + 1.9 \\ &= 3.78 + 1.9 \\ &= 5.68 \end{aligned}\right\} \text{Prediction}$$

$$\left.\begin{aligned} e_1 &= 3.16 - 3 \\ &= 0.16 \\ e_2 &= 5.68 - 6 \\ &= -0.32 \end{aligned}\right\} \text{Errors}$$

Gradient for m

$$\frac{\partial J}{\partial m} = (0.16)(1) + (-0.32)(3)$$
$$= 0.16 - 0.96$$
$$= -0.80$$

Gradient for b

$$\partial J = 0.16 + (-0.32)$$
$$= -0.16$$

Then the update

$$m_3 = 1.26 - 0.1(-0.80)$$
$$= 1.26 + 0.08$$
$$= 1.34$$

$$b_3 = 1.9 - 0.1(-0.16)$$
$$= 1.9 + 0.016$$
$$= 1.916$$

Results After 3 iteration

$m = 1.34$

$b = 1.916$

CamScanner

Iteration 4

$m_3 = 1.34$

$b_3 = 1.916$

$$\left.\begin{aligned} \hat{y}_1 &= 1.34(1) + 1.916 \\ &= 3.256 \\ \hat{y}_2 &= 1.34(3) + 1.916 \\ &= 4.02 + 1.916 \\ &= 5.936 \end{aligned}\right\} \text{Prediction}$$

$$\left.\begin{aligned} e_1 &= 3.256 - 3 \\ &= 0.256 \end{aligned}, \quad \begin{aligned} e_2 &= 5.936 - 6 \\ &= -0.064 \end{aligned}\right\} \text{Errors}$$

Since $\frac{2}{n} = \frac{2}{2} = 1$

<u>Gradient for m</u>

$$\frac{\partial J}{\partial m} = (0.256)(1) + (-0.064)(3)$$

$$= 0.256 - 0.192$$

$$= 0.064$$

Gradient for b

$$\frac{\partial J}{\partial b} = 0.256 + (-0.064)$$
$$= 0.192$$

Update for Iteration 4

$$m_4 = 1.34 - 0.1(0.064)$$
$$= 1.34 - 0.0064$$
$$= 1.3336$$

$$b_4 = 1.916 - 0.1(0.192)$$
$$= 1.916 - 0.0192$$
$$= 1.8968$$

Results from Iteration 4

$m_4 = 1.3336$

$b_4 = 1.8968$

x y